

Name : Hafsa Tariq

Registration Number: FA24B1-SE-071

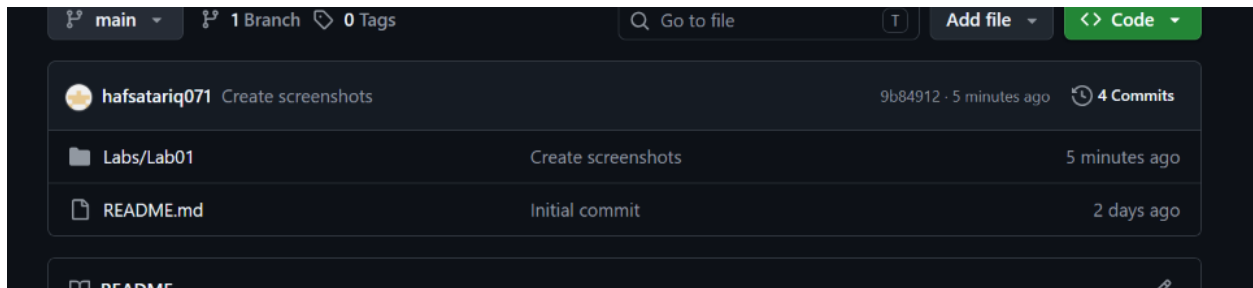
Course Code: BSE-665

Section: B

GitHub Username : hafsatarig071

Lab #1

Getting started:



Demonstrates the required Labs/Lab01/screenshots/ directory structure created inside the repository.

Taskk#1: create and configure your github account

Email*

hafsatariqslb@gmail.com

Password*

.....



Password should be at least 15 characters OR at least 8 characters including a number and a lowercase letter.

Username*

hafsatariq071



Username may only contain alphanumeric characters or single hyphens, and cannot begin or end with a hyphen.

Your Country/Region*

Pakistan



For compliance reasons, we're required to collect country information to send you occasional updates and announcements.

Email preferences

☒ Receive occasional product updates and announcements

Create account >

Public profile

Name

hafsatariq071

Your name may appear around GitHub where you contribute or are mentioned. You can remove it at any time.

Public email

Select a verified email to display

You can manage verified email addresses in your [email settings](#)

Bio

Tell us a little bit about yourself

You can @mention other users and organizations to link to them.

Pronouns

she/her

URL

Profile picture

Edit

Demonstrates the created public GitHub profile displaying the username hafsatariq071

1

General

Owner *

 hafsatarig071

Repository name *

CC

✔ CC is available.

Great repository names are short and memorable. How about [sturdy-parakeet?](#)

Description

cloud computing lab

19 / 350 characters

2

Configuration

Choose visibility *

Choose who can see and commit to this repository

 Public

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

On ☒

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore

Add license

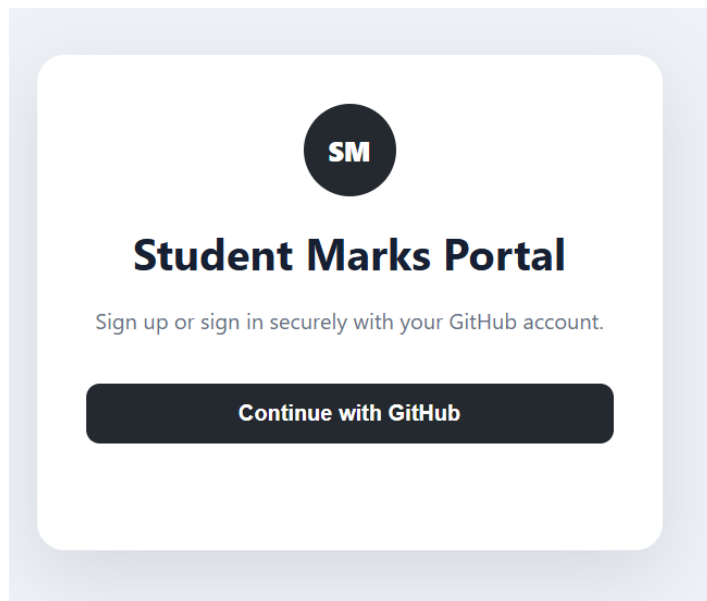
Licenses explain how others can use your code. [About licenses](#)

No license

Create repository

Demonstrates create public repository

Taskk#2: register in the student marks portal



ID

Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number

2024-BSE-71

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course

SE-665 — Cloud Computing

Section

Section B

Submit enrollment

Sign out

Demonstrates student GitHub profile information and registration details integrated inside the portal.

STUDENT MARKS PORTAL

Course enrollments

Sign out

hafsatriq071

2024-BSE-71

+ Enroll in another course

ENROLLMENT STATUS

Your courses

Cloud Computing (SE-665) Section B

Waiting for administrator approval

Pending approval

Demonstrates successful submission of course enrollment awaiting approval inside the Student Marks Portal.

Task#3:Download the VMware workstation pro and ubuntu server

Connecting to Support Portal ECX



Broadcom Inc. Customer sign-in

Username

Enter your username

☐ Remember me

Next

User Registration

Create your Account

Email Address

hafsatarigislb@gmail.com



JFQV

Verified Successfully

Next

⚠ Use of a shared email account or a distribution list (DL) is a security risk and a violation of the

Register for VMware workstation pro

←

Verify your Email Address

Enter the verification code sent to
hafs***lb@gmail.com**

7 2 9 3 9 3

Didn't receive the code? [Resend](#)

Having trouble? [Click here](#) for help.

Verify & Continue

Verifying email address

←

Complete your Registration

* First Name
Hafsa

* Last Name
Tariq

* Country
Pakistan

Job Title
Engineer

* Password

* Confirm Password

✓ Password matched

☒ I accept the [Terms of Use](#). Your privacy is important to us. Please see our [Privacy Policy](#) for more information.

Create Account

Completing the profile

BROADCOM®

Broadcom Inc. Customer sign-In

Username
Hafsa Tariq

☐ Remember me

Next

Entering username and password

Connecting to AH EXT PROD

BROADCOM®

Not Hafsa Tariq?

Password
.....

[Forgot Password?](#)

Back Sign In

Trade Compliance and Download Conditions

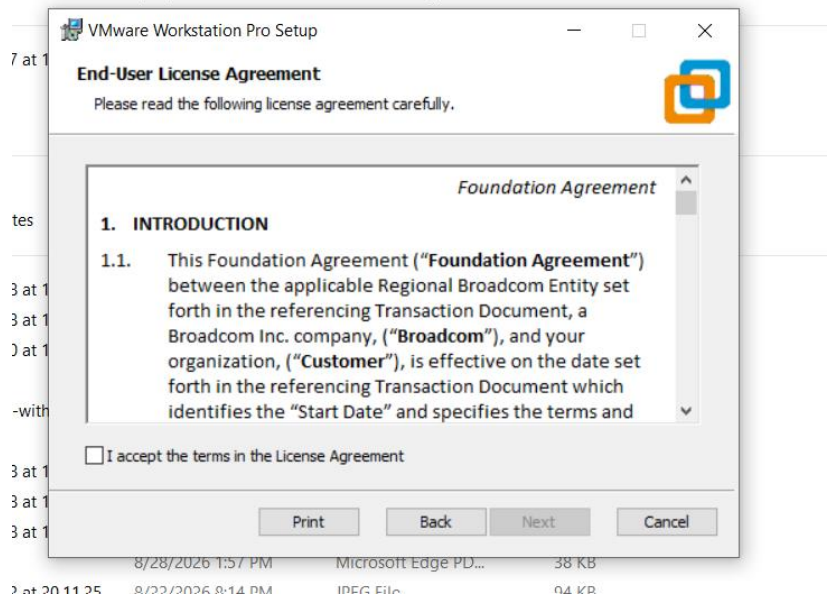
First Name: Hafsa Last Name: Tariq Email: hafsatariq5tb@gmail.com * Company: Company
 * Address1: Address1 Address2: Address2 * City: Islamabad * State/Province: punjab
 * Country: PAKISTAN * Zip/Postal Code: 44000

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By selecting "I Agree" and submitting my information, I acknowledge and agree (i) to the export requirements specified above, (ii) that any download I access via this site is only permitted for the internal use and benefit of the originally licensed company, which must hold a valid license for the relevant Broadcom Software, (iii) that any download I access via this site may only be used within such company's facilities, and (iv) that all such downloads will not be disseminated to any other company.

☒ I Agree ☐ I Do Not Agree

Cancel Submit



Ubuntu 26.04.1 LTS



The latest LTS version of Ubuntu Server. LTS stands for long-term support — which means five years of free security and maintenance updates, extended to up to 15 years with [Ubuntu Pro](#).

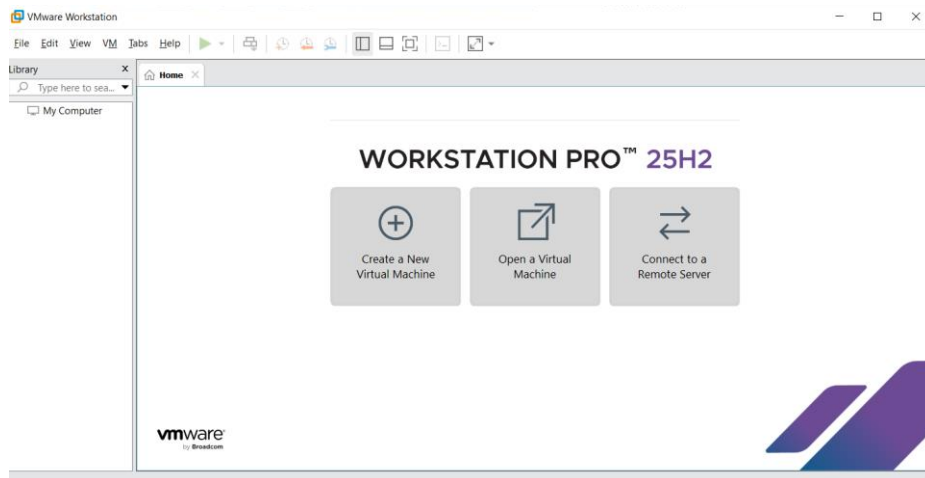
Intel or AMD 64-bit architecture

[Download](#) 3GB

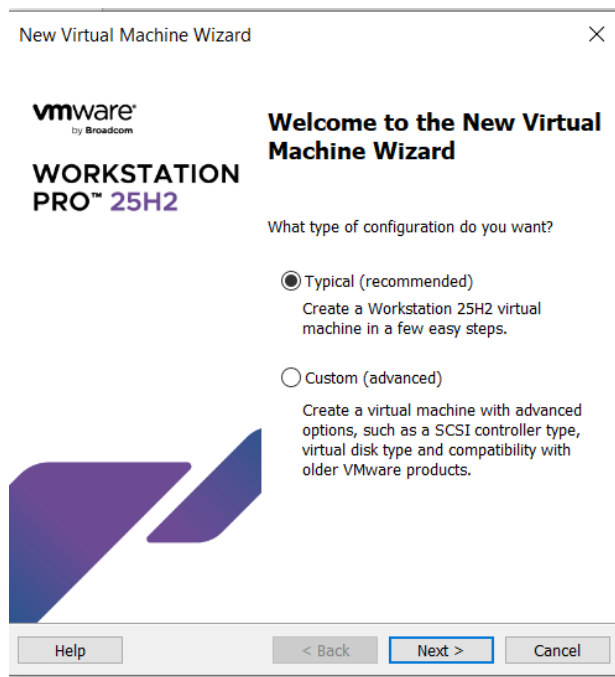
For other versions of Ubuntu Server including torrents, the network installer, a list of local mirrors, and past releases [check out our alternative downloads](#).

Installing ubuntu

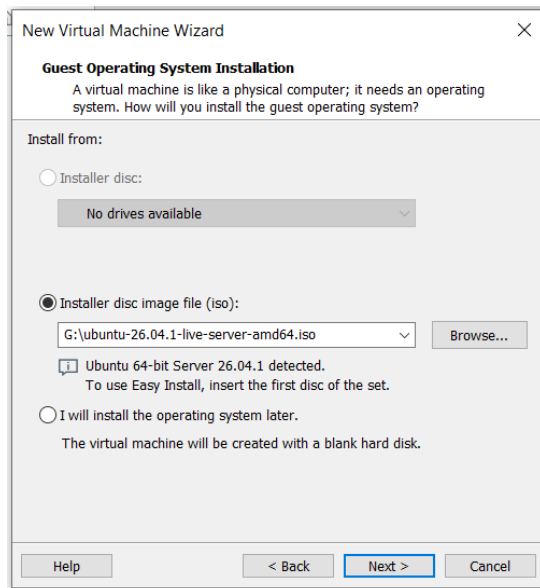
Taskk#4: Craete ubuntu server virual machine



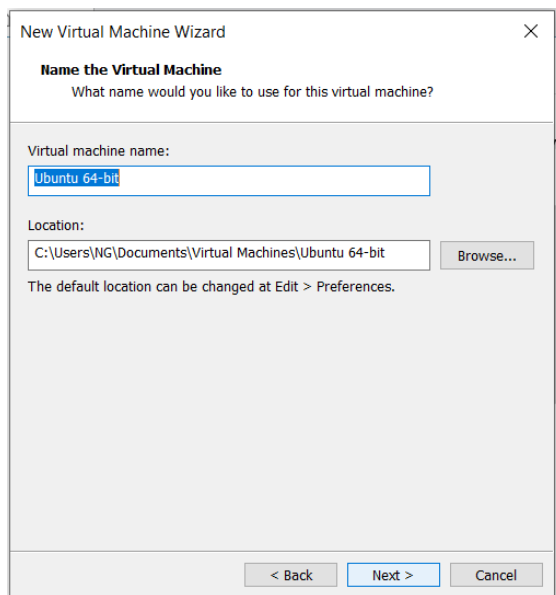
Demonstrates opening the VMware New Virtual Machine Wizard to begin virtual machine setup.

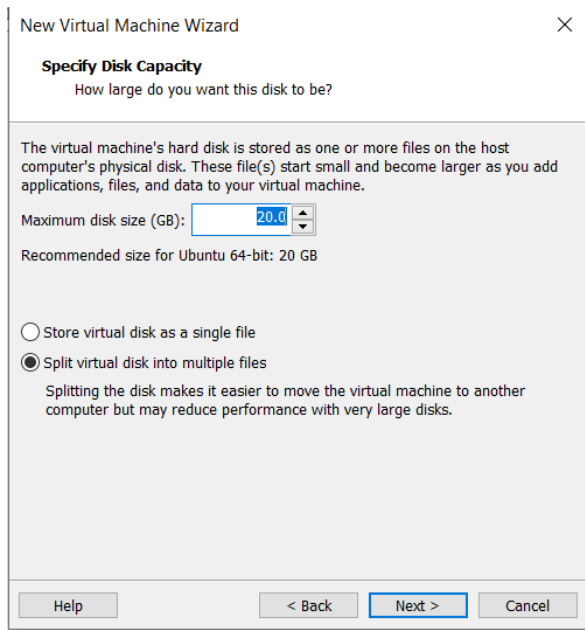


Demonstrates selecting the Typical configuration option for the virtual machine setup.

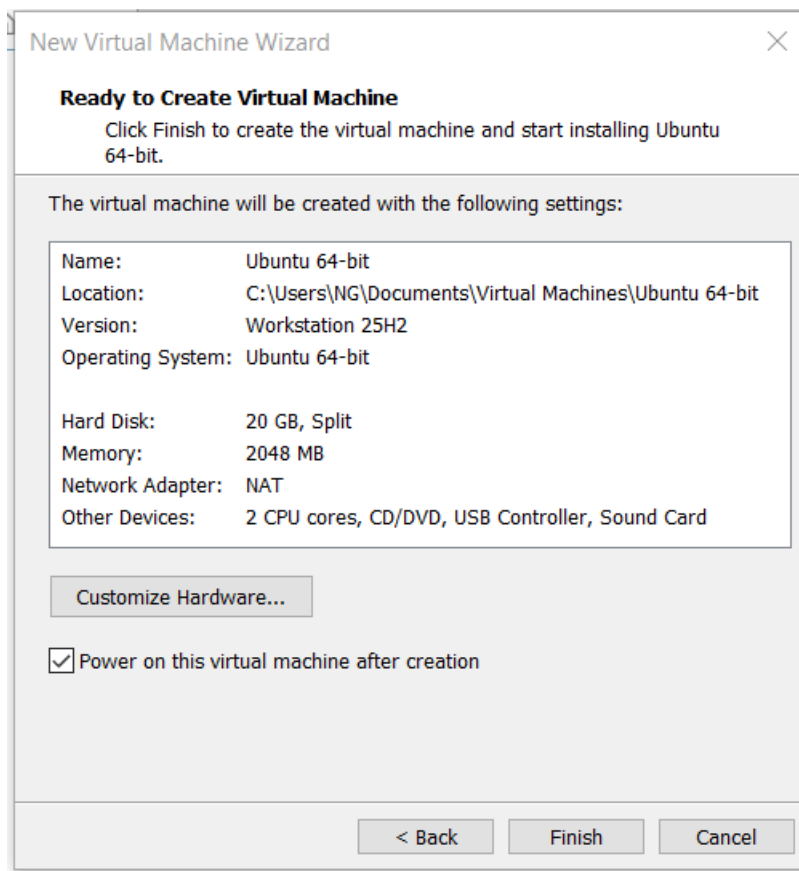


Displays the selected local file path for the Ubuntu Server ISO installer image.





Demonstrates the final virtual machine configuration summary prior to creation.



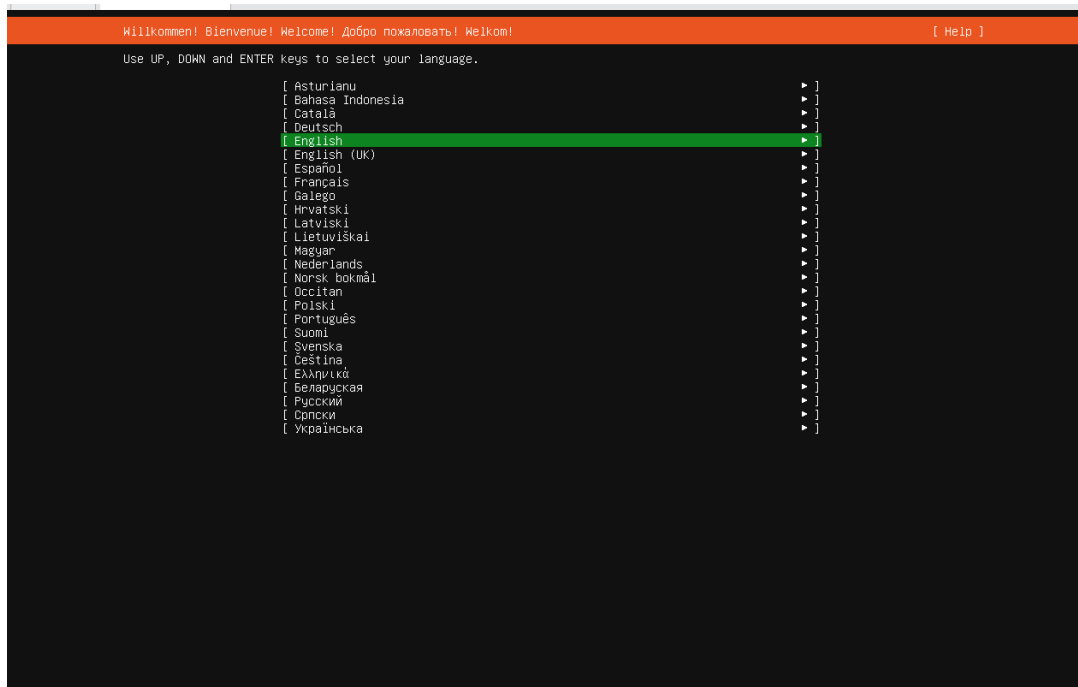
```

Begin: Configuring fstab... .. done.
Begin: Setting up locales... .. done.
Begin: Setting up automatic login... .. done.
Begin: Setting hostname... .. done.
Begin: Setting up console keyboard... .. done.
Begin: Applying desktop settings... .. done.
Begin: Regenerating SSL certificate... .. done.
Begin: Loading preseed file... .. done.
Begin: Adding live session user... .. ^[spassud: password changed.
done.
Begin: Setting up init... .. done.
Begin: Configuring accessibility options... .. done.
Begin: Disabling update-notifier... .. done.
Begin: Enabling detection of crashes... .. done.
Begin: Disabling unnecessary KDE services... .. done.
Begin: Disabling the lock screen for LXQt (LP: #2000787)... .. done.
Begin: Generating APT cache for cdrom ... Ign:1 file:/cdrom resolute InRelease
Get:2 file:/cdrom resolute Release [664 B]
Get:2 file:/cdrom resolute Release [664 B]
Get:3 file:/cdrom resolute Release.gpg [228 B]
Get:3 file:/cdrom resolute Release.gpg [228 B]
Get:4 file:/cdrom resolute/main amd64 Packages [39.1 kB]
Get:5 file:/cdrom resolute/restricted amd64 Packages [13.9 kB]

```

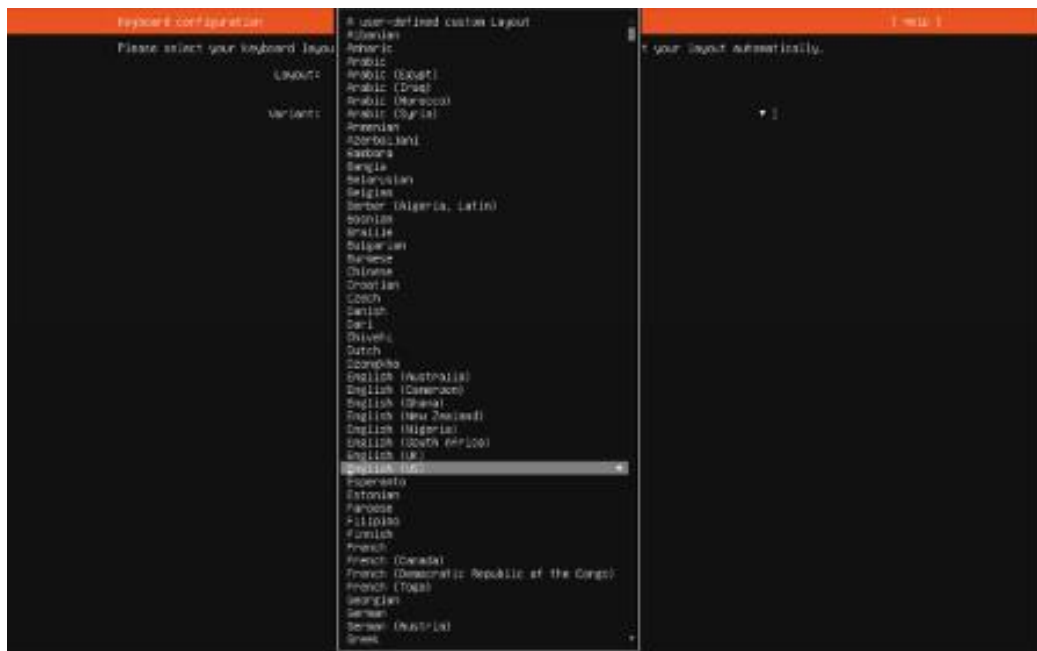
Demonstrates the Ubuntu Server installer boot screen upon powering on the virtual machine.

Taskk#5: choose the installation language

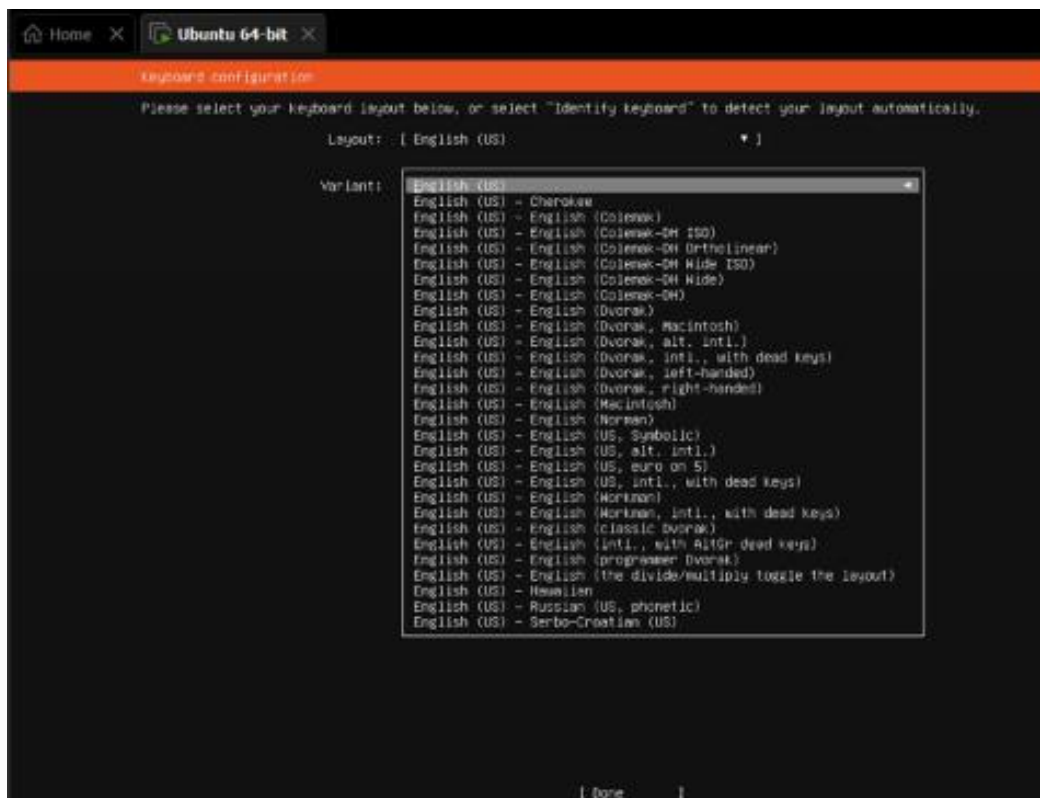


Demonstrates selecting English as the primary installer language.

Taskk#6: configure the keyboard layout

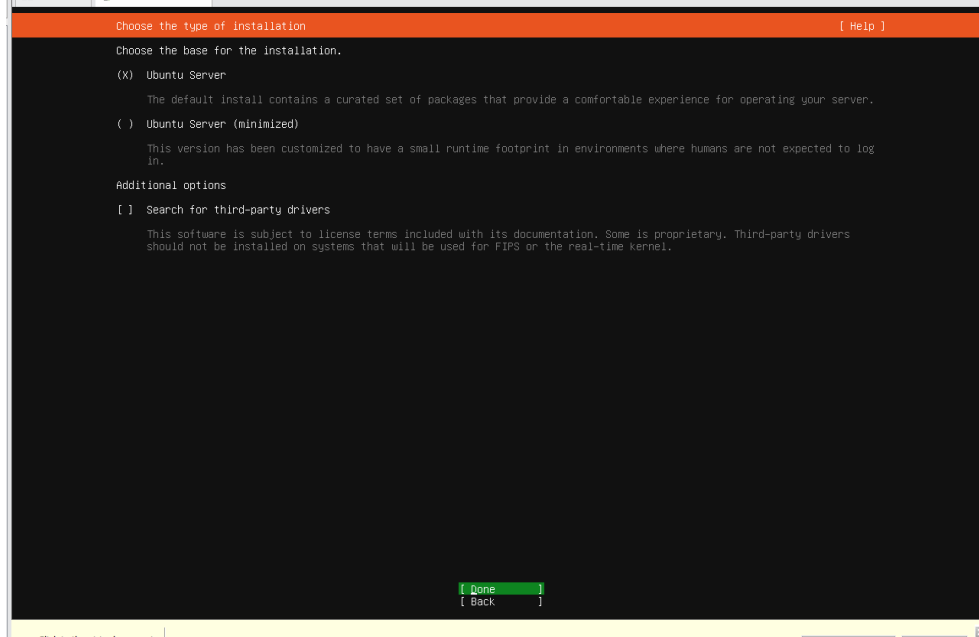


Demonstrates selecting the primary keyboard layout (English US) during the Ubuntu Server installation setup.



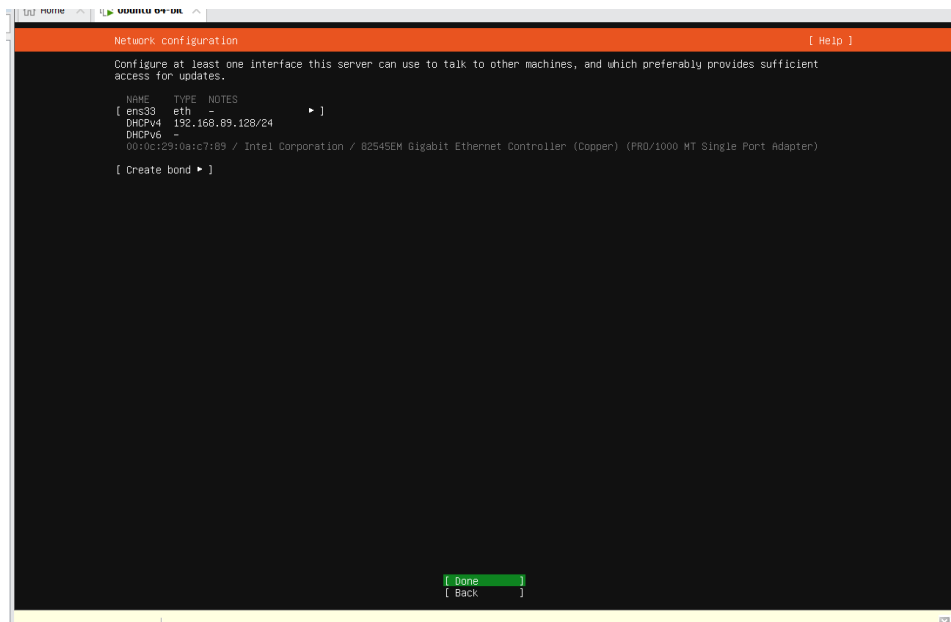
Displays selecting the matching keyboard layout variant (English US) before proceeding with system setup.

Taskk#7: choose the ubuntu server installation type

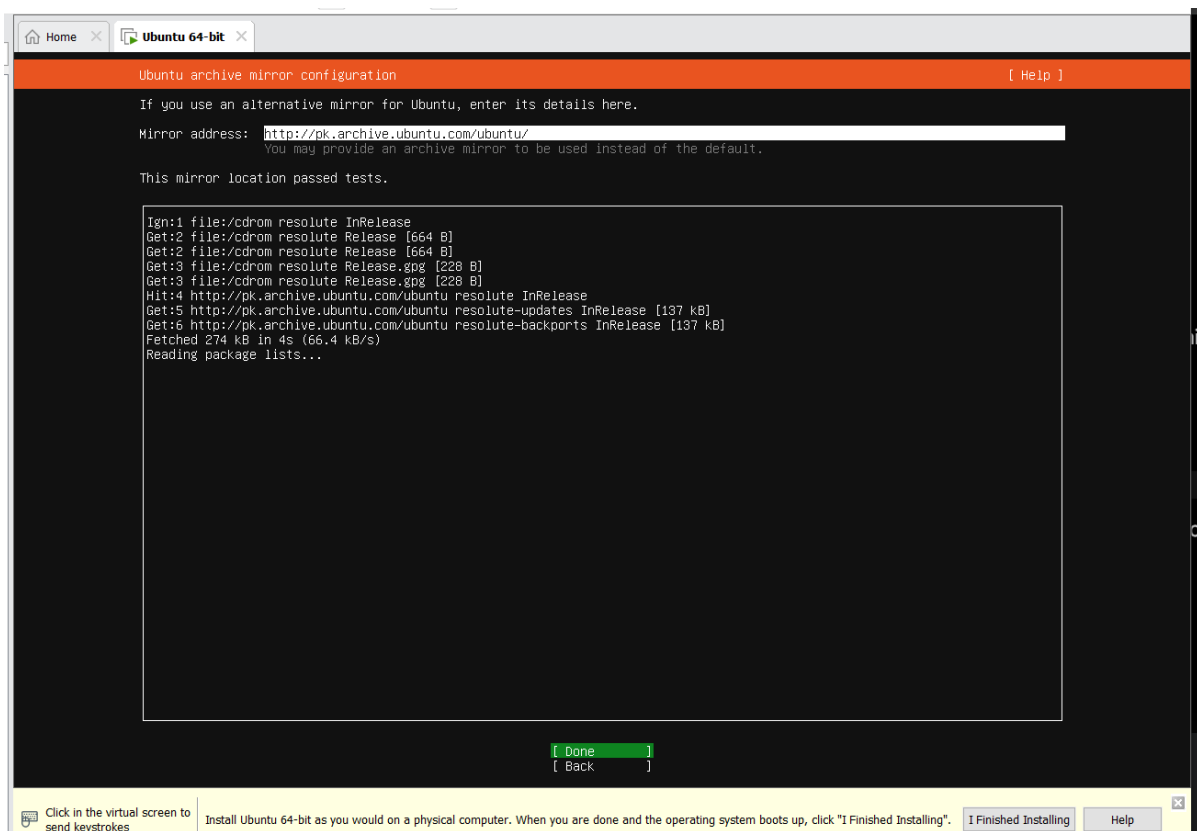
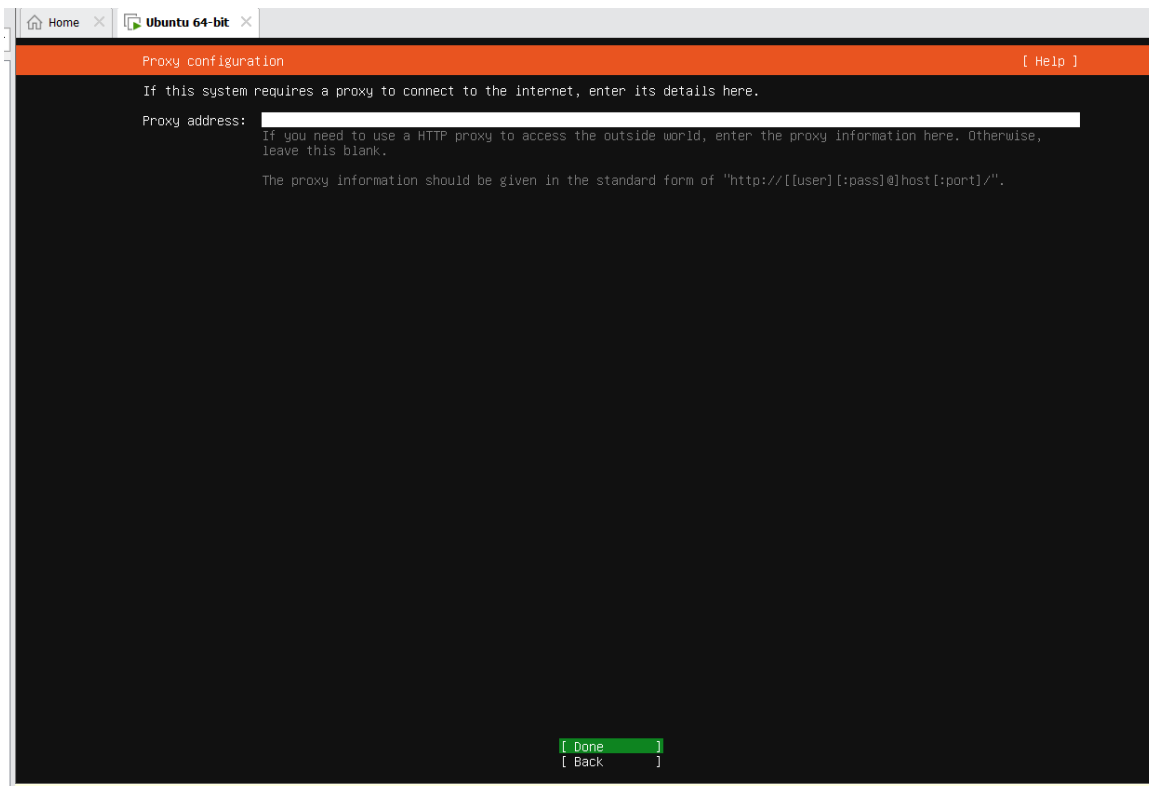


Demonstrates selecting the standard base Ubuntu Server installation option instead of MAAS deployment.

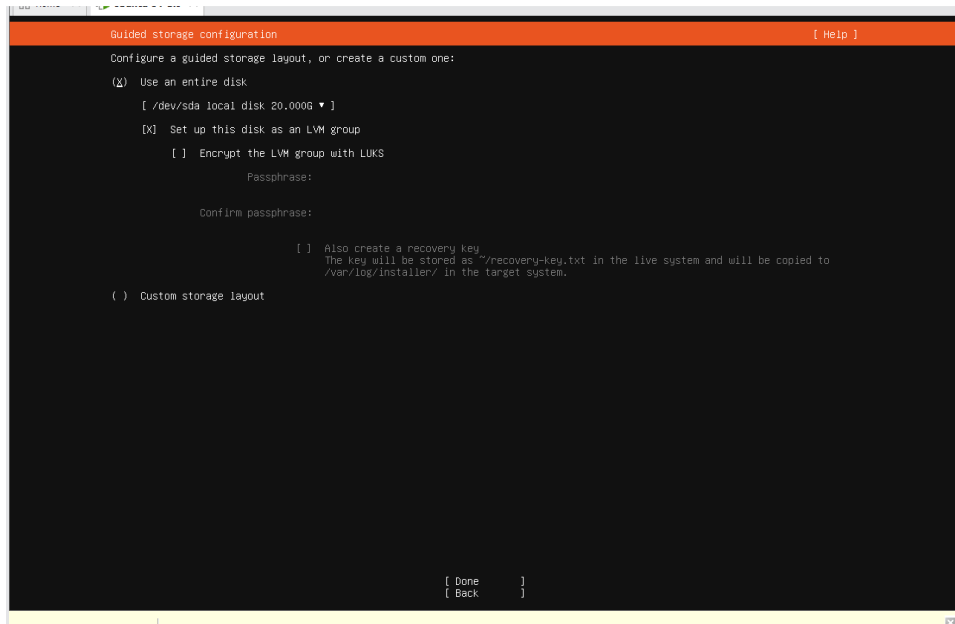
Taskk#8:configure the network



Demonstrates detection of the primary Ethernet network interface (ens33) by the Ubuntu Server installer.

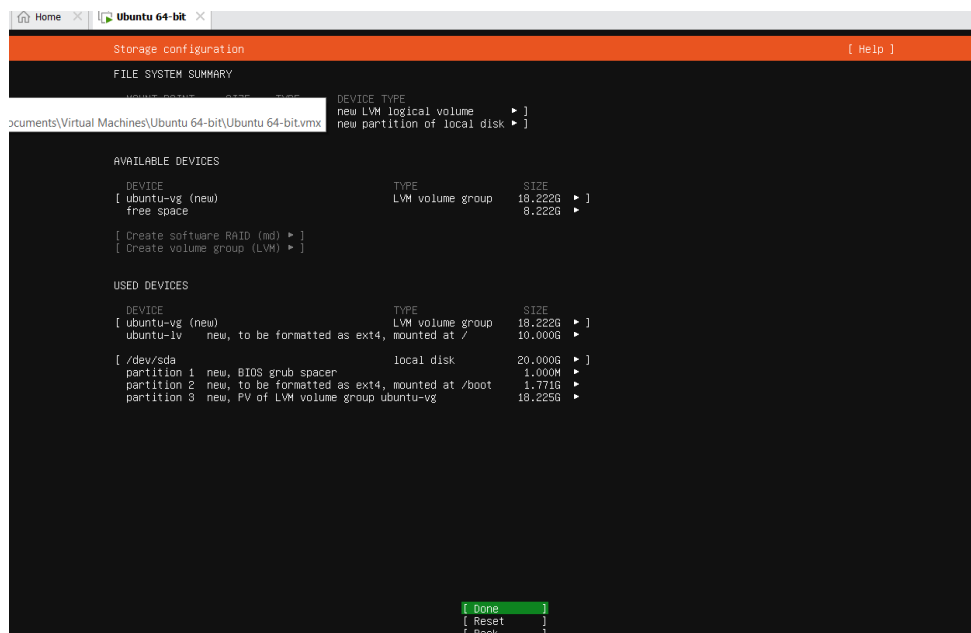


Task#9:configure storage



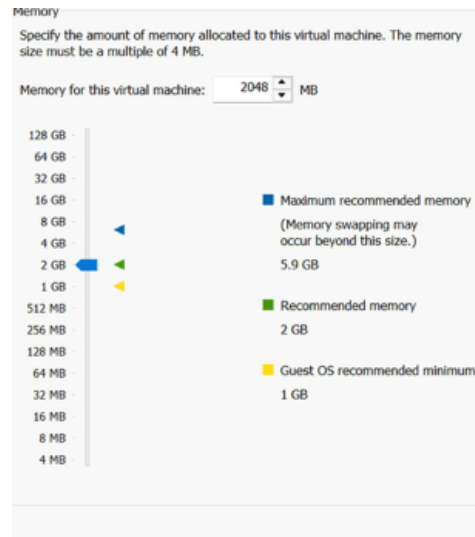
Demonstrates selecting the guided storage option to use the allocated VMware virtual disk for installation.

Task 10: Select the Installation Disk



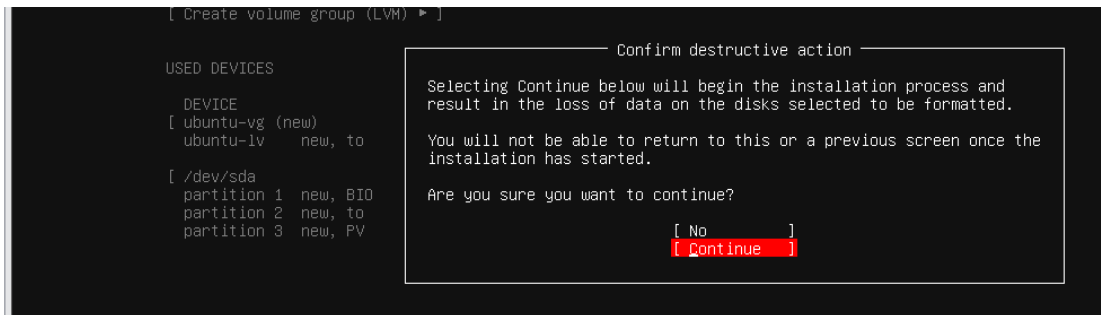
Displays the reviewed filesystem storage layout and partition summary prior to applying changes.

Task#11: review partition



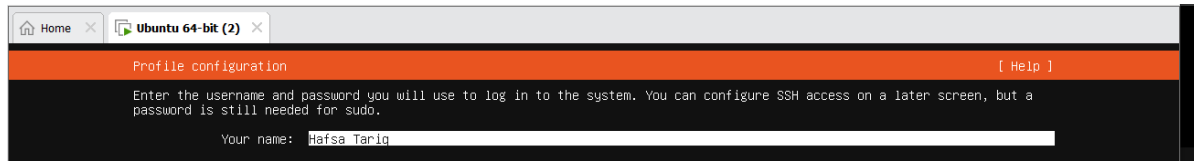
Demonstrates the proposed partition layout and filesystem summary on the target VMware virtual disk.

Task #12: confirm the storage changes

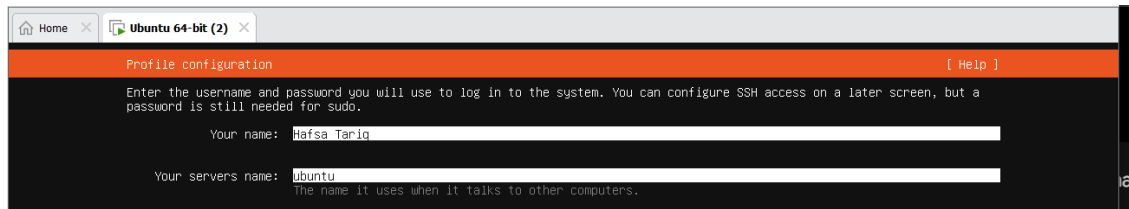


Displays the installer's destructive-action warning prompt prior to overwriting the virtual disk.

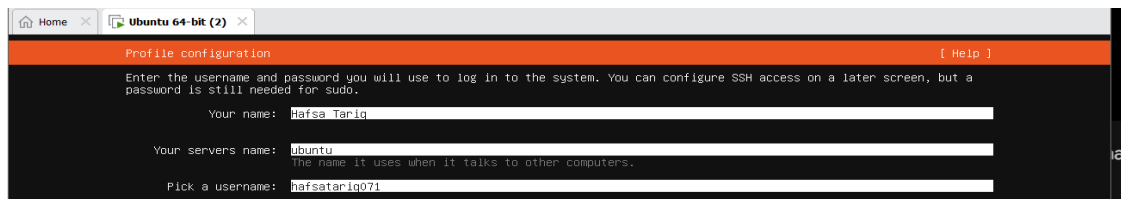
Task #13: configure the ubuntu profile



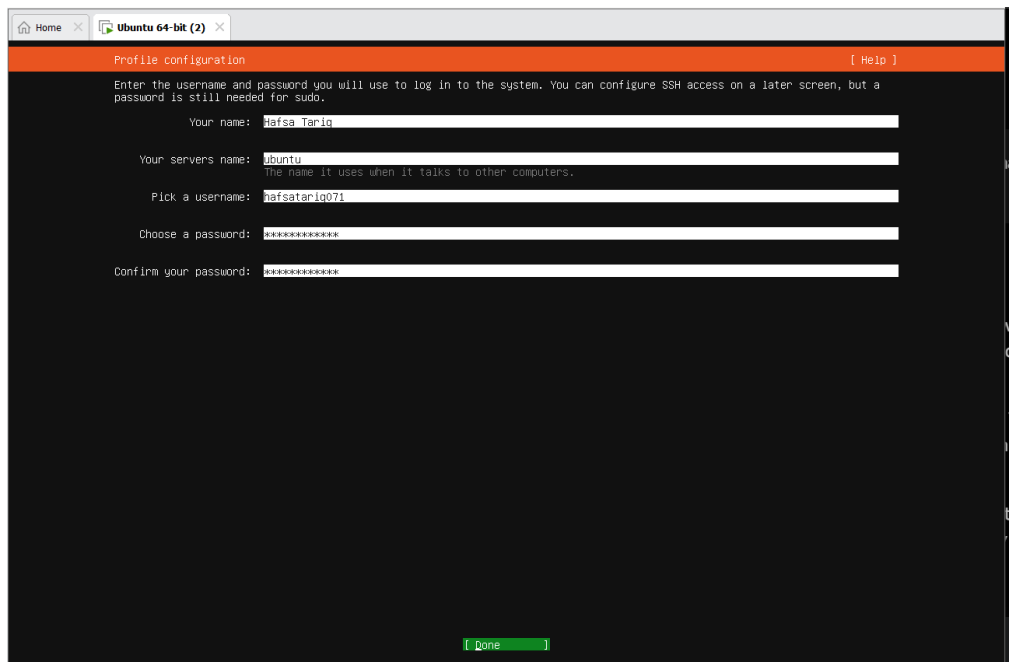
Demonstrates entering the student's full actual name in the profile configuration screen.



Demonstrates setting the server hostname to ubuntu during system configuration.

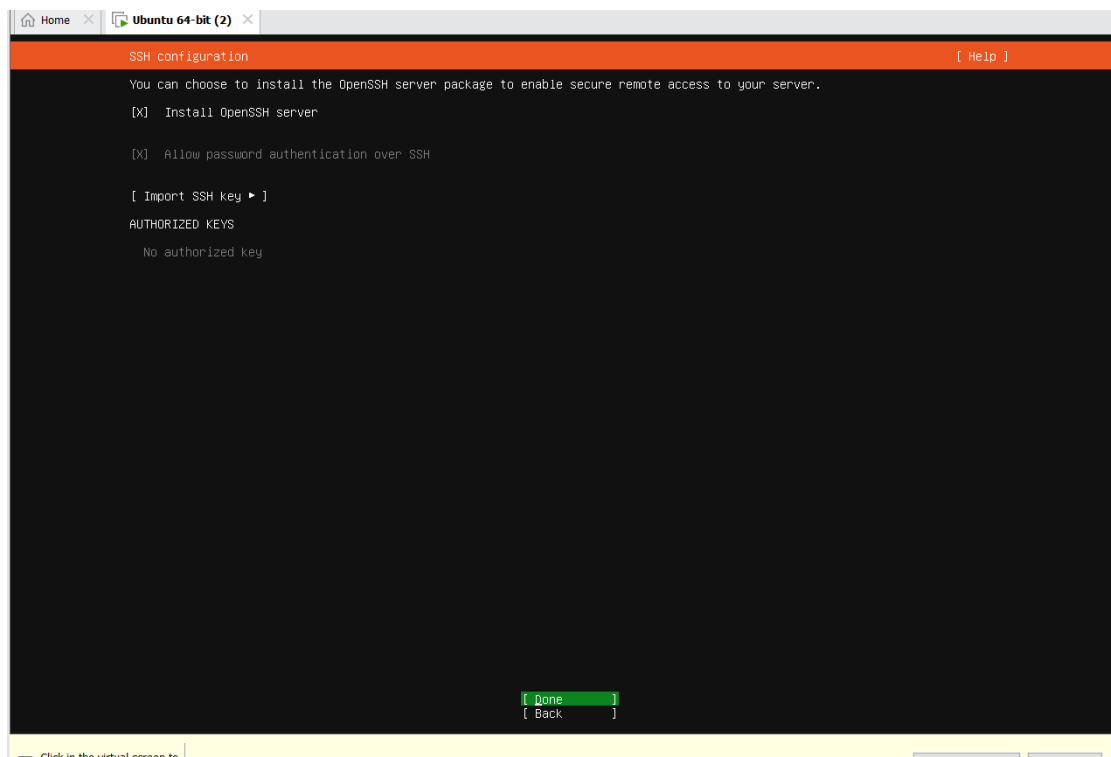
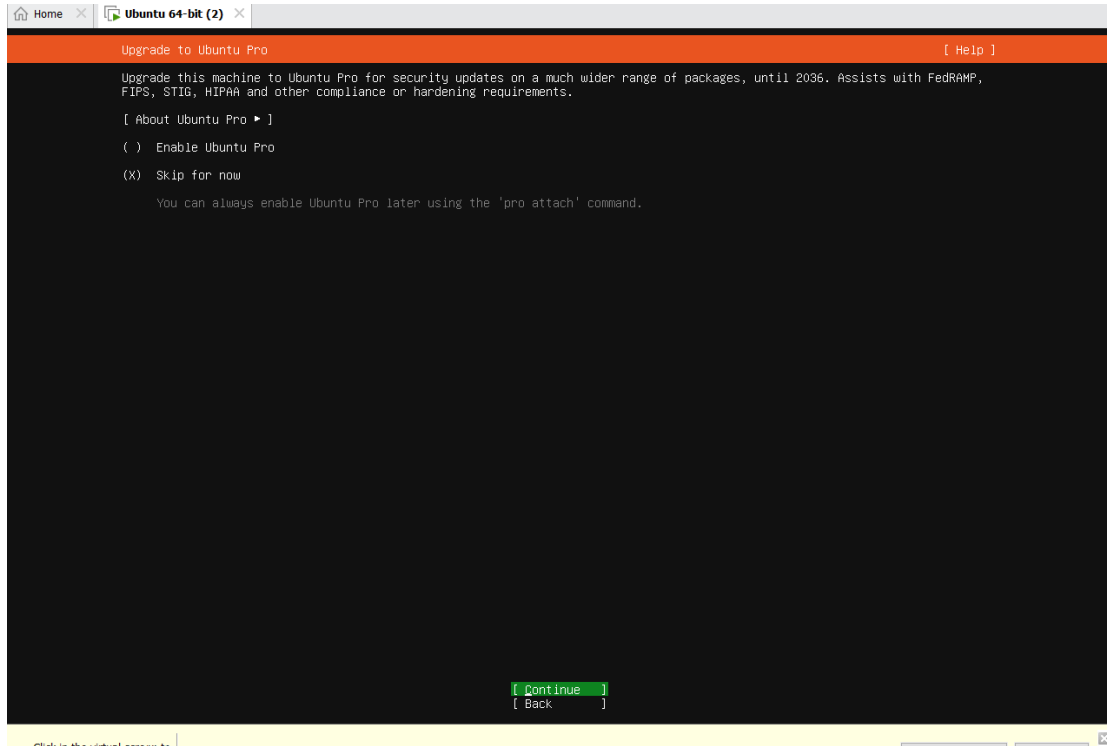


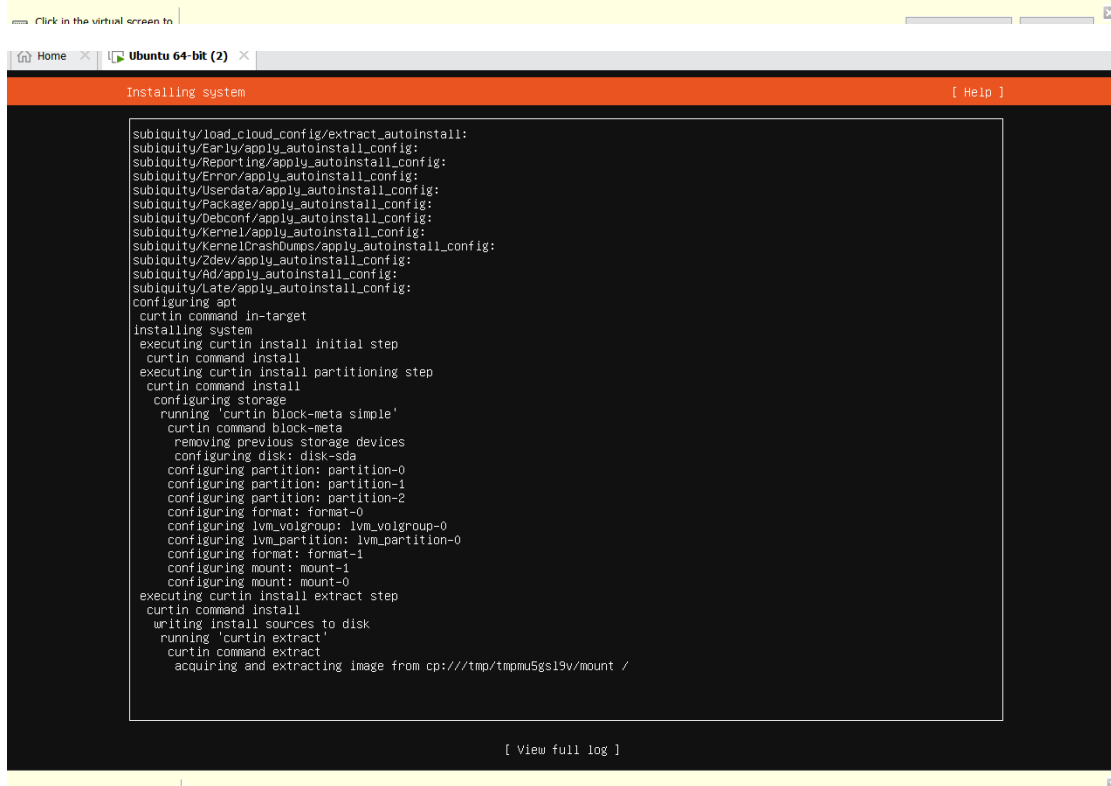
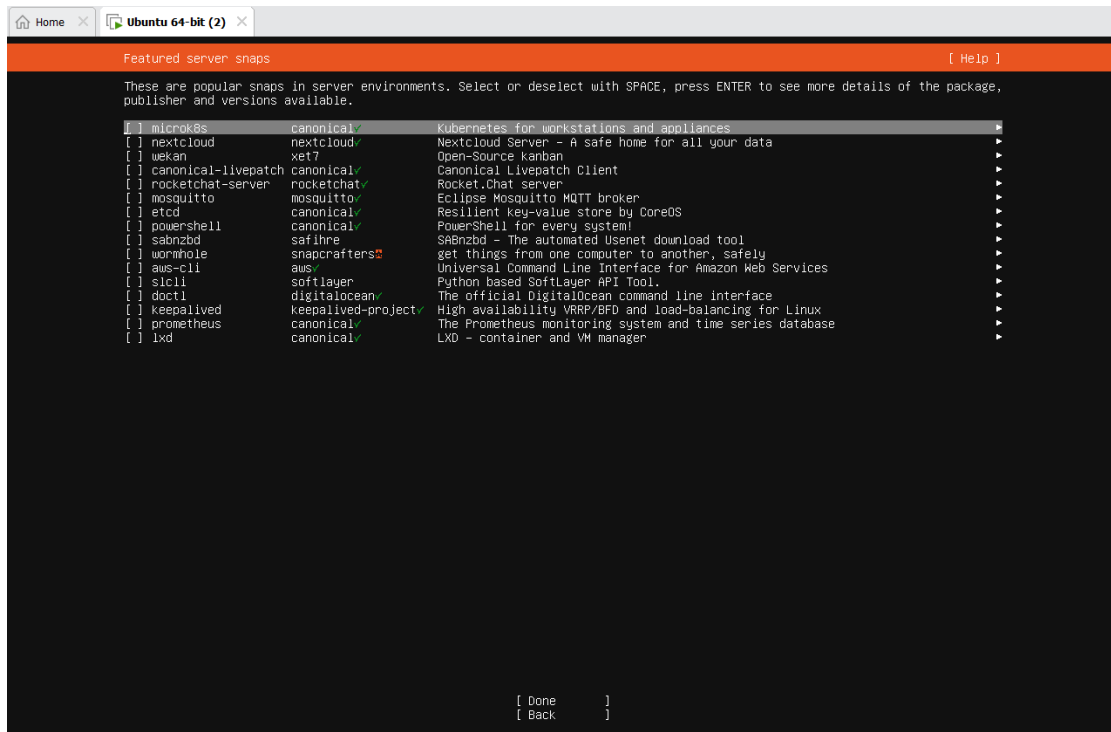
Demonstrates configuring the primary system username to match the student's GitHub handle



Demonstrates the fully completed profile setup page featuring hostname and user credentials.

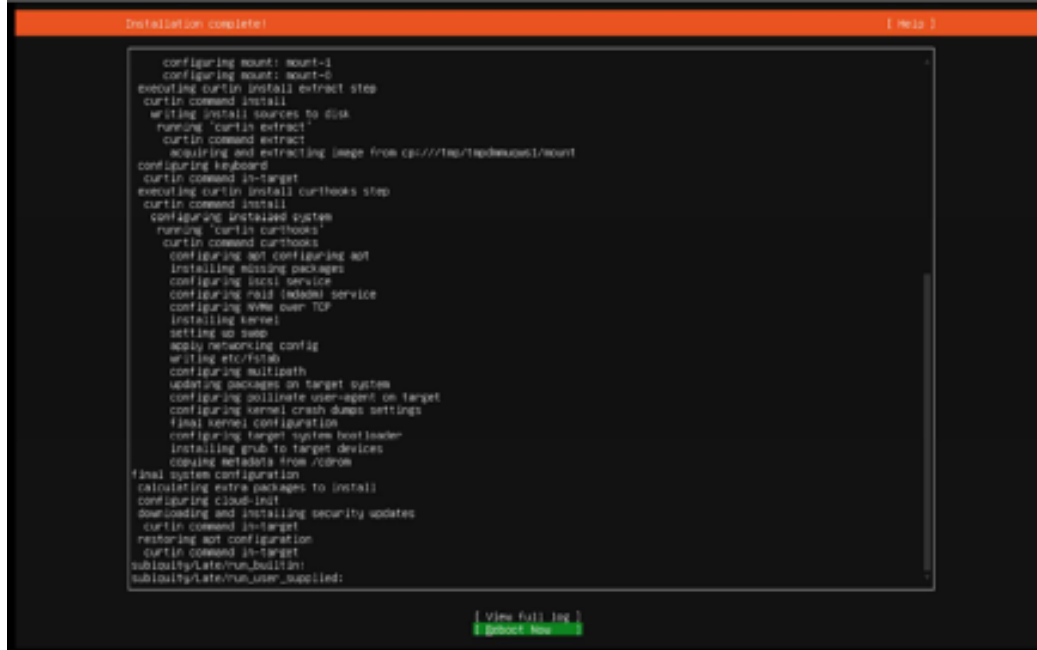
Task #14: complete the software installation





Displays the ongoing system installation progress and package extraction on the virtual machine.

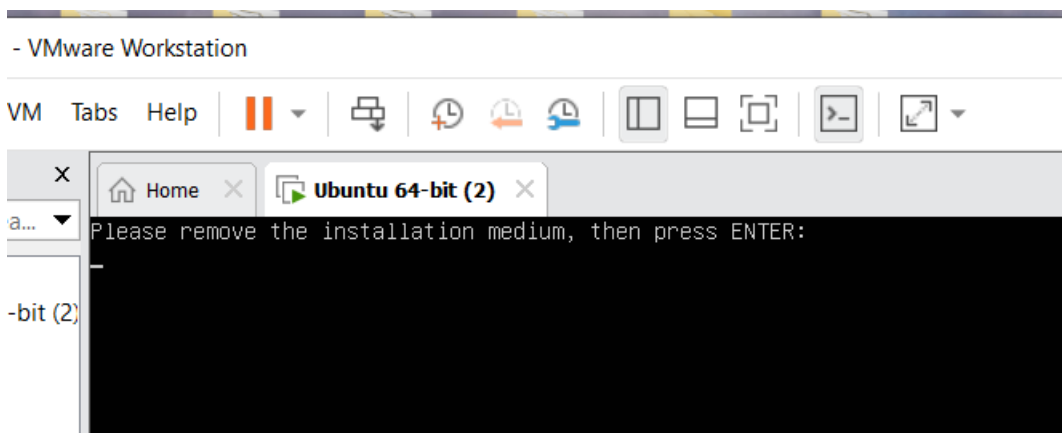
Task #15: Reboot and verify the ubuntu identity



```
Installation complete! [ Help ]

configuring mount: mount-1
configuring mount: mount-2
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cpi:///ftp/debian/ubuntu
configuring keyboard
curtin command in-target
executing curtin install curthooks stop
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring base service
configuring raid (mdadm) service
configuring VMWare over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring polkit user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquity/late/mus_builtins:
subiquity/late/mus_user: supplied:
```

Demonstrates the installation complete notification prior to system reboot.



```
[ 51.752625] cloud-init[1146]: +-----[SHA256]-----+
^[s
ubuntu login:
ubuntu login:
ubuntu login: hafsatarig071
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep 16 06:02:12 PM UTC 2026

System load:  0.4           Processes:           257
Usage of /:   50.5% of 9.75GB Users logged in:       0
Memory usage: 21%          IPv4 address for ens33: 192.168.89.131
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

hafsatarig071@ubuntu:~$
```

Demonstrates successful local terminal login using user credentials

Task#16 find ubuntu server ip address

```
hafsatarig071@ubuntu:~$ ls
hafsatarig071@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:9a:e1:8d brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c299ae18d
    inet 192.168.89.131/24 metric 100 brd 192.168.89.255 scope global dynamic ens33
        valid_lft 1516sec preferred_lft 1516sec
    inet6 fe80::20c:29ff:fe9a:e18d/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
hafsatarig071@ubuntu:~$ hostnamectl
  Static hostname: ubuntu
    Icon name: computer-vm
    Chassis: vm
Chassis Asset Tag: No Asset Tag
  Machine ID: 94800ab9ae734f06b88f456e4165b5b9
    Boot ID: 40d19c3a205d41bf9007db738b664f4c
  AF_VSOCK CID: 396026253
  Virtualization: vmware
  Operating System: Ubuntu 26.04.1 LTS
    Kernel: Linux 7.0.0-31-generic
    Architecture: x86_64
  Hardware Vendor: VMware, Inc.
  Hardware Model: VMware Virtual Platform
  Hardware Version: None
  Firmware Version: 6.00
    Firmware Date: Mon 2025-03-24
    Firmware Age: 1y 5month 3w 3d
hafsatarig071@ubuntu:~$ df -h
Filesystem              Size  Used Avail Use% Mounted on
tmpfs                   319M  1.3M  318M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv 9.8G  5.0G  4.4G  54% /
tmpfs                   798M    0  798M   0% /dev/shm
none                    1.0M    0   1.0M   0% /run/credentials/systemd-journald.service
tmpfs                   798M    0  798M   0% /tmp
none                    1.0M    0   1.0M   0% /run/credentials/systemd-resolved.service
/dev/sda2               1.8G  190M  1.5G  12% /boot
none                    1.0M    0   1.0M   0% /run/credentials/systemd-networkd.service
none                    1.0M    0   1.0M   0% /run/credentials/getty@tty1.service
tmpfs                   160M   8.0K  160M   1% /run/user/1000
hafsatarig071@ubuntu:~$
```

Demonstrates executing the ip addr command inside the Ubuntu Server terminal.

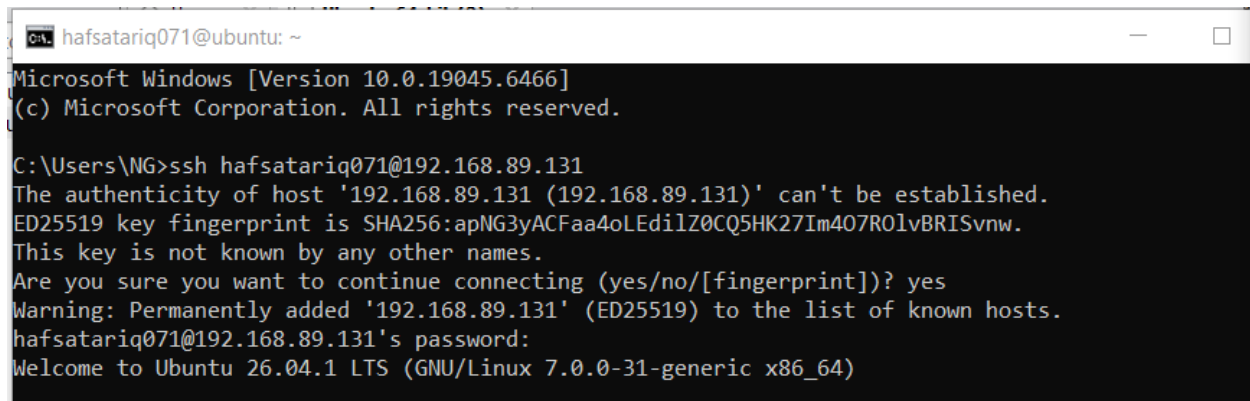
Displays the assigned network interface IPv4 address

Task 17: Connect to Ubuntu Server from Windows

```
C:\Users\NG>ssh hafsatarig071@192.168.89.131
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\NG>ssh hafsatarig071@192.168.89.131
The authenticity of host '192.168.89.131 (192.168.89.131)' can't be established.
ED25519 key fingerprint is SHA256:apNG3yACFaa4oLEdILZ0CQ5HK27Im407R0lvBRISvnm.
This key is not known by any other names.
```

Demonstrates initiating an SSH remote connection from Windows terminal to the Ubuntu Server.

A screenshot of a Windows terminal window. The title bar shows 'hafsatariq071@ubuntu: ~'. The terminal content shows the Windows command prompt with the command 'ssh hafsatariq071@192.168.89.131'. It displays the host's authenticity warning, the user's confirmation 'yes', the warning to add the host to the known hosts list, the password prompt, and the final 'Welcome to Ubuntu' message.

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\NG>ssh hafsatariq071@192.168.89.131
The authenticity of host '192.168.89.131 (192.168.89.131)' can't be established.
ED25519 key fingerprint is SHA256:apNG3yACFaa4oLEdilZ0CQ5HK27Im407R0lvBRISvnw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.89.131' (ED25519) to the list of known hosts.
hafsatariq071@192.168.89.131's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)
```

Displays accepting the host key fingerprint to establish a trusted SSH connection

```
hafsatarig071@ubuntu: ~  
Microsoft Windows [Version 10.0.19045.6466]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\NG>ssh hafsatarig071@192.168.89.131  
The authenticity of host '192.168.89.131 (192.168.89.131)' can't be established.  
ED25519 key fingerprint is SHA256:apNG3yACFaa4oLEdilZ0CQ5HK27Im407ROlvBRISvnw.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.89.131' (ED25519) to the list of known hosts.  
hafsatarig071@192.168.89.131's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Fri Sep 18 04:36:35 PM UTC 2026  
  
System load:  0.1              Processes:            223  
Usage of /:   50.7% of 9.75GB   Users logged in:     0  
Memory usage: 22%              IPv4 address for ens33: 192.168.89.131  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
12 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
Last login: Fri Sep 18 16:30:00 2026 from 192.168.89.131
```

Demonstrates successful authentication and remote terminal login to Ubuntu Server over SSH.

```
hafsatarig071@ubuntu: ~  
hafsatarig071@ubuntu:~$
```

Displays the remote shell prompt `hafsatariq071@ubuntu:~$` confirming active SSH session identity.